HOTEL BOOKING DEMAND ANALYTICS

Statistical Summary Report

This report summarizes the statistical analysis performed on the Hotel Booking Demand dataset. The analysis includes assumption testing, statistical test selection, hypothesis testing, business conclusions, and recommendations.

1. Assumption Testing Summary

Assumption	Statistical Test	Test Statistic	P-value	Decision
Normality	Shapiro-Wilk	0.9846	0.0	Reject Normality
Homogeneity of Variance	Levene Test	3113.0008	0.0	Reject Equal Variance

The Shapiro-Wilk test indicated that the dataset is not normally distributed. Levene's test confirmed unequal variances. Based on these results, suitable parametric and non-parametric statistical methods were selected.

2. Statistical Test Selection

Research Objective	Statistical Test	Reason for Selection
Compare ADR between Hotel Types	Independent t-Test	Compare the means of two independent groups.
Compare ADR across Booking Seasons	One-Way ANOVA	Compare the means of more than two groups.
Measure relationship between Lead Time and ADR	Pearson Correlation	Measure the strength of a linear relationship.
Measure association between Hotel Type and Cancellation	Chi-Square Test	Determine the association between categorical variables.
Compare Lead Time between Cancelled and Non-Cancelled Bookings	Mann–Whitney U Test	Compare two independent groups when normality is violated.
Compare ADR across Customer Types	Kruskal–Wallis Test	Compare more than two independent groups when assumptions are violated.
Measure monotonic relationship between Lead Time and Special Requests	Spearman Correlation	Measure the strength of a monotonic relationship between variables.

Statistical tests were selected based on research objectives, variable types, and assumption testing results.

3. Hypothesis Testing Outcomes

Statistical Test	Test Statistic	P-value	Decision
Independent t-Test	$t = 55.8138$	0.0000	Reject H_0
One-Way ANOVA	$F = 1857.6878$	0.0000	Reject H_0
Pearson Correlation	$r = 0.0383$	$7.31 \times 10^{\blacksquare 2 \blacksquare}$	Reject H_0
Chi-Square Test	$\chi^2 = 551.9297$	$4.79 \times 10^{\blacksquare 122}$	Reject H_0
Mann–Whitney U Test	$U = 1.108985e+09$	0.0000	Reject H_0
Kruskal–Wallis Test	$H = 3938.9729$	0.0000	Reject H_0
Spearman Correlation	$\rho = 0.1109$	$4.83 \times 10^{\blacksquare 23 \blacksquare}$	Reject H_0

Seven hypothesis tests were performed. All tests produced p-values below the significance level of 0.05. Therefore, all null hypotheses were rejected, confirming statistically significant results.

4. Statistical Decision Summary

Decision Criterion	Result
Significance Level (α)	0.05
Confidence Level	95%
Number of Tests Performed	7
Tests with $p < 0.05$	7
Overall Statistical Decision	Reject All Null Hypotheses

5. Business Conclusions

Business Area	Conclusion
Booking Behaviour	Booking patterns vary significantly across different hotel types and seasons.
Customer Behaviour	Customer characteristics influence booking behaviour and stay preferences.
Cancellation Analysis	Booking cancellations are significantly associated with hotel type, market segment, and lead time.
Revenue Analysis	Average Daily Rate (ADR) differs across hotel types and seasons, affecting revenue generation.
Seasonal Trends	Seasonal demand influences occupancy levels and pricing strategies.
Statistical Analysis	All hypothesis tests confirmed statistically significant relationships within the dataset.

6. Key Recommendations

Recommendation	Expected Benefit
Reduce Booking Cancellations	Lower revenue loss caused by cancellations.
Improve Revenue Management	Increase profitability through optimized room pricing.
Optimize Seasonal Pricing	Maximize occupancy during peak and off-peak seasons.
Increase Direct Bookings	Reduce dependency on third-party booking channels.
Enhance Customer Experience	Improve customer satisfaction and guest retention.
Use Predictive Analytics	Forecast demand and support strategic decision-making.

Final Conclusion: The statistical analysis provides evidence-based insights for improving hotel revenue management, reducing cancellations, optimizing seasonal pricing, increasing direct bookings, and enhancing customer satisfaction.